2.2 The Verlet Method

The Verlet method (also known as leapfrog or Störmer-Verlet) is a second order method that is popular for molecular simulation. It is specialized to problems that can be expressed in the form $\dot{\boldsymbol{q}} = \boldsymbol{v}, \boldsymbol{M}\dot{\boldsymbol{v}} = \boldsymbol{F}(\boldsymbol{q})$, with even dimensional phase space $\mathbb{R}^{2N}$, which includes constant energy molecular dynamics. Some generalizations exist for other classes of Hamiltonian systems.

The Verlet method is a numerical method that respects certain conservation principles associated to the continuous time ordinary differential equations, i.e. it is a geometric integrator. Maintaining these conservation properties is essential in molecular simulation as they play a key role in maintaining the physical environment. As a prelude to a more general discussion of this topic, we demonstrate here that it is possible to derive the Verlet method from the variational principle. This is not the case for every convergent numerical method. The Verlet method is thus a special type of numerical method that provides a discrete model for classical mechanics.

2.2.1 Hamilton's Principle and Equations of Motion

Hamilton's principle of least action provides a mechanism for deriving equations of motion from a Lagrangian. Recall from Chap. 1 that the Lagrangian for the N-body system is defined by

$$L(\boldsymbol{q}, \boldsymbol{v}) \stackrel{\text{def}}{=} \frac{\boldsymbol{v}^T \boldsymbol{M} \boldsymbol{v}}{2} - U(\boldsymbol{q}),$$

where $\boldsymbol{M}$ is the mass matrix and the potential energy function U is, for simplicity, taken to be smooth (C^2). We consider the collection of all twice continuously differentiable curves in the configuration space which start from a certain point $\boldsymbol{Q}$ and end at another given point $\boldsymbol{Q}'$. We may think of any such curve as being represented by a parameterization $\boldsymbol{q}(t)$, $t \in [\alpha, \beta]$ with $\boldsymbol{q}(\alpha) = \boldsymbol{Q}$ and $\boldsymbol{q}(\beta) = \boldsymbol{Q}'$, where the components of $\boldsymbol{q}(t)$ are C^∞ functions. Denote by $G = G(\boldsymbol{Q}, \boldsymbol{Q}', \alpha, \beta)$ the class of smooth parameterized curves such that $\boldsymbol{q}(\alpha) = \boldsymbol{Q}, \boldsymbol{q}(\beta) = \boldsymbol{Q}'$. Then the *classical action* (or, simply, *action*) of the Lagrangian L is defined for any $\Gamma \in G$ by

$$\mathcal{A}_L(\Gamma) \stackrel{\text{def}}{=} \int_\alpha^\beta L(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) \mathrm{d}t.$$

The variational calculus approach to classical mechanics is based on minimizing the action $\mathcal{A}_L$ over the class G of parameterized curves. This is normally referred to as the "principle of least action". It is difficult to provide a physical motivation for this concept, but it is normally taken as a foundation stone for classical mechanics.